

Experimental Investigation of Airfoil Aerodynamics and Flow Control Concepts: Mono and Biplane Configurations

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Abstract—Although airfoils are one of the most fundamental components of aerospace engineering, relatively little is known about their aerodynamic interactions in biplane configurations, as most studies tend to solely focus on monoplanes. This study addresses this gap by analyzing airfoils in both mono and biplane configurations. The study focuses on four different airfoils: NACA 0012, 2212, 2412, and 2612. For initial analysis, 2D simulations of these airfoils were conducted in XFLR5 and ANSYS Fluent Flow. Following the theoretical analysis, models were 3D printed and tested in a wind tunnel at a Reynolds number of $Re = 20100$. First, CFD (computational fluid dynamic) simulations were conducted for a theoretical analysis which was followed by wind tunnel testing of 3D printed models for an experimental analysis. 51 biplane configurations were tested in combinations determined by airfoil shape and decalage test matrices. Furthermore, finite-span aspect-ratio corrections were applied to the CFD results to distinguish 3D downwash effects from the behavior of low-Reynolds-number viscous flow. The experimental results demonstrated that aerodynamic efficiency for airfoil shape optimization is influenced more through changes in lift rather than in drag. The best performing airfoil shape combination was further optimized through the decalage test matrix. Finally, the highest performing configuration was determined to be NACA 2212 on the top at 0° angle of attack and NACA 2212 on the bottom at 5° angle of attack. The most efficient combinations generally used a converging wing combination that accelerated airflow through the gap, creating a concentrated Venturi effect, while negative lower-wing angles caused noticeable losses in lift and an overall worse performance. These findings emphasize the importance of airfoil selection and decalage in determining the aerodynamic efficiency of low-Reynolds-number biplane configurations, considerably

contributing to the broader study of this field.

I. INTRODUCTION

The idea of biplanes (two wings stacked one above the other) has been considered and studied ever since the inception of airplanes. The main motivation behind early biplanes was to reduce weight and to provide structural support for the wings [1]. Over time they lost popularity in favor of monoplanes (one-winged airplanes). Using CFD simulations, airfoils were modeled and simulated to graph the streamlines and velocity contours. After the simulations, the airfoils of interest were modeled using Computer-Aided Design (CAD) software and 3D-printed for testing in a wind tunnel. The motivation to focus on biplanes, rather than monoplanes, stemmed from the limited modern research in the field. Biplanes can be useful alternatives to monoplanes in many applications due to their distinct behavior and structure, such as for crop dusting and landing on short airfields, partly due to their greater maneuverability, shorter wingspans, and higher lift at low speeds. A computational and experimental analysis and optimization of factors, such as stagger, decalage, gap, and airfoil shape, has the potential to reveal new insights significant to the field of aerospace engineering as a whole. The novelty is not just a source of the possible applications, but a supplement for the lack of understanding in biplane aerodynamics which can provide deeper understanding of

countless physical phenomena and aspects crucial to aerospace engineering [2].

II. BACKGROUND

A. Airfoils

An airfoil is a two-dimensional cross-section of a wing. They are among the most essential components of aerospace engineering. Airfoils are constructed around three primary measurements: the leading and trailing edges, the chord line, and the camber line. The NACA naming convention identifies airfoils based on a four digit code: the first digit represents the maximum camber (m) as a percentage of the chord length, the second digit represents max camber location (p) as a fraction of ten on the chord line, and the final two digits represent the maximum thickness (t) of the airfoil as a percentage of the chord.

B. Aerodynamic Coefficients and Parameters

1) *Angle of Attack*: Angle of Attack will be denoted by (α) and is defined as the angle formed between a wing's chord line and the oncoming relative wind (U_∞) [3] [4].

2) *Coefficient of Lift and Drag*: Lift (L) and drag (D) forces are converted into dimensionless coefficients, C_L and C_D , respectively, and are created using dynamic pressure (q) and planform area (A)[5]. The following are equations for both coefficients.

$$C_L = \frac{L}{\frac{1}{2}\rho U^2 A} = \frac{L}{qA} \quad (1)$$

$$C_D = \frac{D}{\frac{1}{2}\rho U^2 A} = \frac{D}{qA} \quad (2)$$

C. Biplanes

Biplanes can initially be simplified as two line segments to solve for airflow and later mathematically generalized to real airfoil shapes. The flow around a cambered airfoil that has zero circulation is the section flow. Physically, however, this would result in an infinite theoretical velocity at the trailing edge. In reality, a pure section flow is never produced since infinite velocity is not physically possible. Instead, a "circulation flow" is superposed on top, which also has an infinite velocity at the trailing edge, but of opposite sign so it cancels with the other infinity. For the biplane, this condition must be satisfied simultaneously at both wings' trailing edges, which can be accomplished by mapping the geometry into a simplified "tandem" arrangement and solving for the necessary circulation strength. Therefore, the influence between two such lines correctly replicates the aerodynamics of two cambered airfoils. The resulting equation for biplane lift is:

$$C_L = 2\pi \sin \beta B + 2\pi \sin \beta_0 \sqrt{B} \quad (3)$$

The angle of attack for a biplane is denoted by β , while β_0 is the effective angle representing the airfoil's camber. Additionally, B is the factor that represents the loss of lift in biplanes (which is always less than 1). The first term

is representative of α -driven lift, and the second term is representative of camber-driven lift. [2]

For upper and lower wings with different amounts of camber, each wing has its own effective-curvature angle, but the total lift given by the equation above only depends on the average of the two. Similarly, for small decalage, the average of the two wings' α can be used to determine the coefficient of lift. In theory, decalage does not significantly influence total lift. However, it does not account for drag, which plays a critical role in overall aerodynamic efficiency.

III. EXPERIMENTAL PROCEDURE

A. Simulations

Each individual airfoil tested was first simulated in XFLR5 and ANSYS. 16 airfoils were first tested in XFLR5, which uses an estimate of the Reynolds-averaged Navier-Stokes equations to generate graphs predicting the drag polar and coefficient of lift vs. α . The six highest performing airfoils in XFLR5 were simulated in ANSYS Fluent Flow, a more complex and powerful simulation software. This generated streamline and velocity contour graphics, along with lift and drag coefficient calculations at various angles of attack. A major limitation of both of these software programs is that they only account for 2D simulations and do not consider the effects of downwash and other phenomena that occur in 3D, making wind tunnel testing necessary.

B. Aspect Ratio / Finite Span Correction

Because the ANSYS simulations were two-dimensional while the wind-tunnel models had a finite span, a finite-span correction was applied to the simulated results before they were compared with the experimental data. This helped separate finite-span effects from other experimental differences. The rectangular models had a span of 150 mm and a chord of 30 mm, giving an aspect ratio of $AR = b/c = 5$. For each airfoil, the 2D lift-curve slope, a_0 , was obtained from the linear pre-stall portion of its ANSYS C_l vs. α data. The ANSYS slope was used instead of the ideal value of 2π , because it reflected the low-Reynolds-number conditions of the study. The corrected slope was calculated as

$$a_{3D} = \frac{a_0}{1 + \frac{a_0}{\pi AR e}}$$

using an assumed Oswald efficiency factor of $e = 0.8$ for an unswept rectangular wing. The uncertainty associated with this assumption is discussed in the limitations section. Predicted losses from the pre-stall slopes ranged from 13.1% for NACA 2612 to 22.0% for NACA 2212. Using the ANSYS zero-lift angle α_{L0} , the corrected lift coefficient was approximated by

$$C_{L,3D}(\alpha) = a_{3D}(\alpha - \alpha_{L0}).$$

Angles were converted to radians when the lift-curve slopes were measured in rad^{-1} . Drag was estimated as

$$C_{D,3D} = C_{D,2D} + \frac{C_{L,3D}^2}{\pi AR e}.$$

The corrected values were compared with the wind-tunnel data. Remaining differences could therefore be examined in relation to turbulence, surface roughness, mounting interference, and wingtip leakage near the tunnel walls. The correction was only applied in the linear pre-stall region and was not used to predict stall.

C. Wind Tunnel Procedure

Using CAD software from OnShape, 4 different types of airfoils (NACA 0012, NACA 2212, NACA 2412, NACA 2612) were imported, modified to include a small hole to insert a probe at six different α , modeled, and 3D printed. Angles of attacks were manually calibrated using a precision protractor before testing to ensure that the alignment was accurate across all trials [6].

For the biplane configuration, a bracket was created to hold the two airfoils at a gap of 30mm with no stagger. The hole for the probe was placed at the quarter chord point in the center of the wing span of the airfoil. The quarter-chord point is an approximation of the aerodynamic center of the airfoil. This location is significant since it is the point along the chord in which the pitching moment remains constant, regardless of α [7]. Therefore, it is a reliable reference point to place the probe in order to measure the lift and drag forces acting on the airfoil.

For the experimentation in the wind tunnel, a test matrix was created that tested every possible combination of airfoil shapes at 0° angles of attack. From the most efficient airfoil combination a second test matrix was created to evaluate these combinations at varying angles of attack to determine the best angle of attack and the critical angle of attack for each airfoil combination. Each airfoil configuration in each test matrix was placed into the wind tunnel.

IV. RESULTS AND DISCUSSION

A. Monoplane Results

For all 4 NACA airfoil types tested, graphs of C_L vs. α were created, with the simulation and experimental results plotted. Velocity contour and streamline diagrams were also created by the simulations. The experimental data tends to over-predict lift for α around 0° to 15° & under-predict for α from 0° to -5° . This recurring trend is likely due to the discrepancy between 2D and 3D predictions.

Compared to each other, NACA 2212 dominates at positive angles of attack, but NACA 2612 has the highest overall C_L and tends to dominate after 15° . Despite all of the airfoils having significantly different values than the simulations, the rate at which the C_D or C_L change, as shown in Figure 4, with respect to α remained very similar, showing that the general trend remained for all the airfoils except NACA 2212.

The NACA 2212 airfoil did not stall like predicted in the simulations at 10° . This occurred due to the formation of short laminar separation bubbles on the surface of the airfoil, delaying the stall beyond the range tested and increasing the lift.

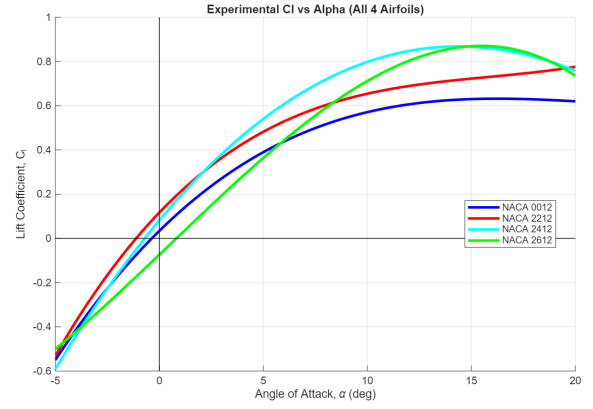


Fig. 1: C_L vs. α for all 4 airfoils overlaid

A reason for the discrepancy and the error could be a result of wall proximity effects. The airfoils were intentionally placed close to the wind tunnel walls to minimize the effect of wingtip vortices forming and downwash. However, there was a 0.5 mm gap between the wingtip and the wall. High-pressure air from below the wing moves around the wingtip to the low-pressure region above it. Since the same amount of air is forced into a smaller gap, the air is accelerated by Bernoulli's principle, which lowers the pressure and pulls more air over the wingtip, decreasing the lift produced.

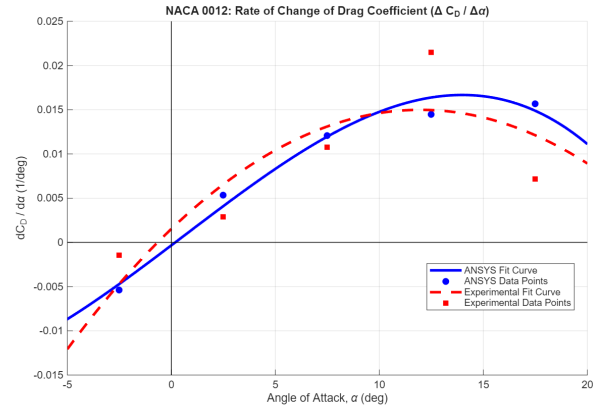


Fig. 2: Rate of Change of Drag Coefficient ($\Delta C_D / \Delta \alpha$)

While the downwash effect was likely still reduced due to the airfoil's proximity to the edge of the wind tunnel, without a perfect seal, wingtip leakage would still occur, affecting the results. This leakage should not be mistaken for wind tunnel wall interference, however.

The true behavior of a wing with the aspect ratio of 5 might not be represented by these results and might end up somewhere in between a model that is fully bound to the walls and a wing in free air due to the leakage around the wingtips.

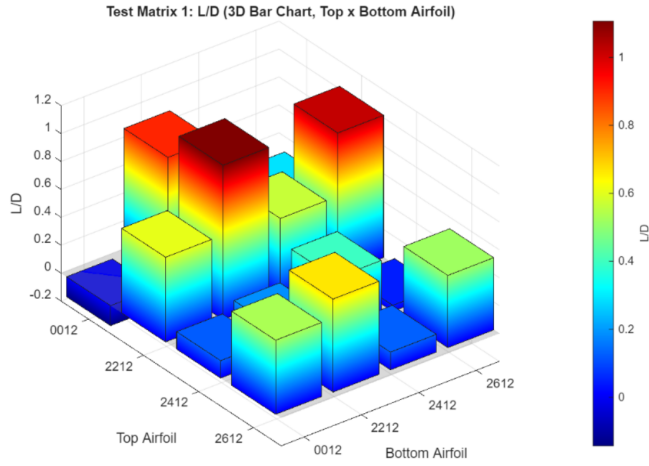


Fig. 3: Test matrix 1: L/D (3D Bar Chart, Top $\times$ Bottom Airfoil)

B. Biplane Wind Tunnel Results

For all four NACA airfoils selected, a test of every possible combination was done: 16 different combinations. The results are demonstrated through the heatmap above. The notation utilized to describe a biplane configuration will be [Top Airfoil 4-digit NACA code] $\times$ [Bottom Airfoil 4-digit NACA code]; ie. 0012 $\times$ 0012. The combination with the highest lift-to-drag ratio is 2212 $\times$ 2212. The lift-to-drag ratio for the combination was 1.1064

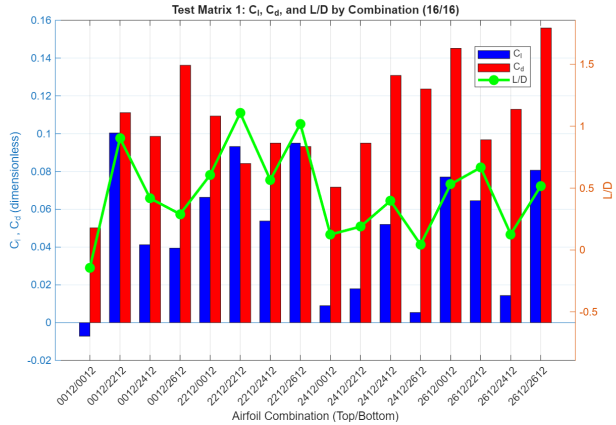


Fig. 4: C_L (blue) and C_D (red) tracked against aerodynamic efficiency (green)

Figure 4 demonstrates that the efficiency ratio is tracked by C_L rather than C_D , as illustrated by the green line following the same trend as the C_L bars. The 2212 $\times$ 2212 combination can optimize camber without causing too much airflow interference between the top and bottom wings. A higher max-camber location combination may have increased lift, but there would be greater interference from the bottom wing, which would cause an overall reduction in lift. It can also be seen through equation 3 that 2212 $\times$ 2212 results in the greatest C_L compared to the other combinations. The

worst performing combination is 0012 $\times$ 0012. As 0012 is the symmetric airfoil, a zero angle of attack does not result in any lift, so all that is measured is drag and interference. All combinations of airfoils, except the 2 most efficient, produced more drag than lift. With 2212 $\times$ 2212 determined as the best configuration, the second test matrix was completed.

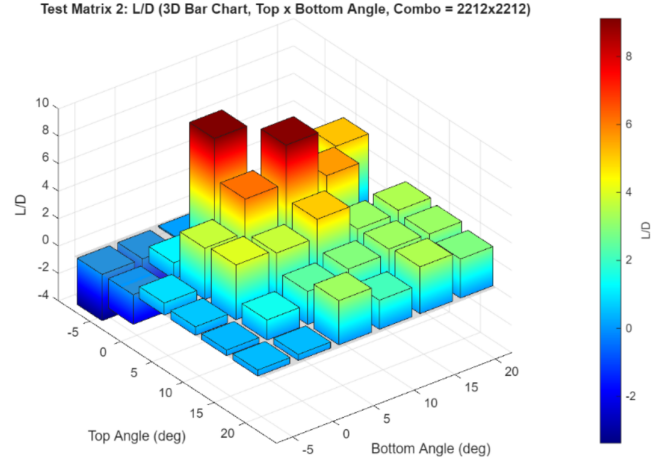


Fig. 5: Test matrix 2: L/D (3D Bar Chart, Top $\times$ Bottom Angle, 2212 $\times$ 2212)

The results of the second matrix are demonstrated in the heatmap above. 36 different combinations were created by varying decalage for the top and bottom airfoils of the 2212 $\times$ 2212 combination. From this, the best lift-to-drag ratio was found. This was done with the top airfoil at 0° and the bottom at 5° which resulted in a ratio of 9.12. The other well performing combinations were 5° $\times$ 10° ($L/D = 8.98$) as well as 5° $\times$ 5° ($L/D = 6.19$).

The three worst-performing decalage combinations did not particularly produce a large amount of drag; rather, the inefficiency came from the reduction in lift force due to earlier flow separation from negative α . Further, equation 3 can be used to verify the same relation, illustrating consistency between experimental and theoretical results. For increased α , the C_L also increases as predicted theoretically, but the C_D also increases, reducing overall efficiency.

1) *Further Discussion on 2212 $\times$ 2212 at 0° $\times$ 5° and 5° $\times$ 10°:* The biplanes at 0° $\times$ 5° and 5° $\times$ 10° performed significantly better than the other combination of angles. They had an L/D ratio of 9.12 and 8.98, respectively. These biplanes both shared a converging geometry on the channel between the airfoils. The local acceleration of the air, due to Bernoulli's principle, most likely contributes to a reshaping of the pressure distribution on the wing surfaces. This creates a more concentrated Venturi effect.

Both airfoils above showed reduced C_D due to this acceleration of air through the gap of the airfoil. This increase in overall efficiency can help to drastically reduce the fuel mass for a given range. Since the wing is more efficient, the aircraft requires less fuel, making it more efficient. This

relationship can be modeled by the Breguet Range Equation, which is conceptually analogous to the famous Tsiolkovsky Rocket equation [8].

2) *Stall-Order Asymmetry*: With the data from the second test matrix, an asymmetrical stall relationship between the two airfoils can be suggested.

The top airfoil stalling has a greater effect on the overall lift than the bottom airfoil stalling. When the bottom airfoil stalls before the top airfoil, the resulting loss of lift is significantly less and depends more on the angle of the top airfoil. All of the best performing biplanes had a greater angle of attack on the bottom airfoils relative to the top airfoil. Meanwhile, all the worst performing biplanes had a top airfoil with the greater angle of attack. The bottom airfoil having a larger angle of attack than the top can help to restore an aircraft in a stall. By one wing stalling before another, it causes a decrease in lift, and allows the aircraft to return to a stable angle of attack since the other airfoil would still be producing lift. With the bottom airfoil being at a higher angle, the overall airfoil performance is increased due to the asymmetry of the stalls and the channel flow effect improving the efficiency of the biplane.

3) *Monoplane and Biplane Configuration Performance Comparison*: For the specific test conditions of the experiments, the optimal biplane (2212×2212 at $0^\circ \times 5^\circ$) performed better by a significant margin compared to the optimal monoplane (2612 at 10°). The L/D was 9.12 and 4.21, respectively, with a difference of 4.91. While promising, more robust testing and consideration of the test-conditions are necessary for certain applications.

C. Limitations

Throughout testing, certain variables were assumed rather than experimentally measured. A turbulence intensity (T.I.) of 0.5% was used in ANSYS simulations instead of the wind tunnel's actual value, which could have contributed to the earlier stall observed experimentally. Because cambered airfoils should generate lift at zero angle of attack, the slightly negative lift measured for the NACA 2412 and NACA 2612 may also reflect sensitivity to T.I.; however, these transient results were retested and verified. An Oswald efficiency factor ($e=0.8$) was also assumed for finite-span corrections. Since e affects the correct lift-curve slope and induced drag, an inaccurate value would change the correction magnitude. Wall proximity and the small wingtip gap may further alter the actual efficiency through confinement and tip-leakage effects. More extensive testing is needed to determine a wing-specific value of e .

V. CONCLUSIONS

This study evaluated the four NACA airfoils, 0012, 2212, 2412, and 2612, in monoplane and biplane configurations using computational analysis and wind-tunnel testing. The NACA 2212×2212 pairing proved most efficient, with the optimal decalage placing the upper wing at 0° and the lower wing at 5° . Overall, the best results paired a small positive upper-wing angle with an equal or greater lower-wing angle,

demonstrating that biplane performance depends on wing-to-wing interaction rather than the behavior of isolated airfoils. Future research should continue by optimizing stagger and gap along with decalage and airfoil shape. This study's findings suggest potential applications in variable-geometry aircraft, where a second wing could improve low-speed performance before being retracted for cruise. Further optimization could be performed at hypersonic speeds, eventually looking to use learned insights to implement the theoretical Busmann Biplane. This study provides a valuable experimental baseline for modern biplane optimization, showing that wing geometry and incidence significantly influence aerodynamic efficiency and support further investigation of biplanes for low-speed and high-lift applications.

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